



DBSCAN ALGORITHM

Subtítulo (introduzir, se aplicável, ou eliminar o campo)

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Dataset1 DBSCAN

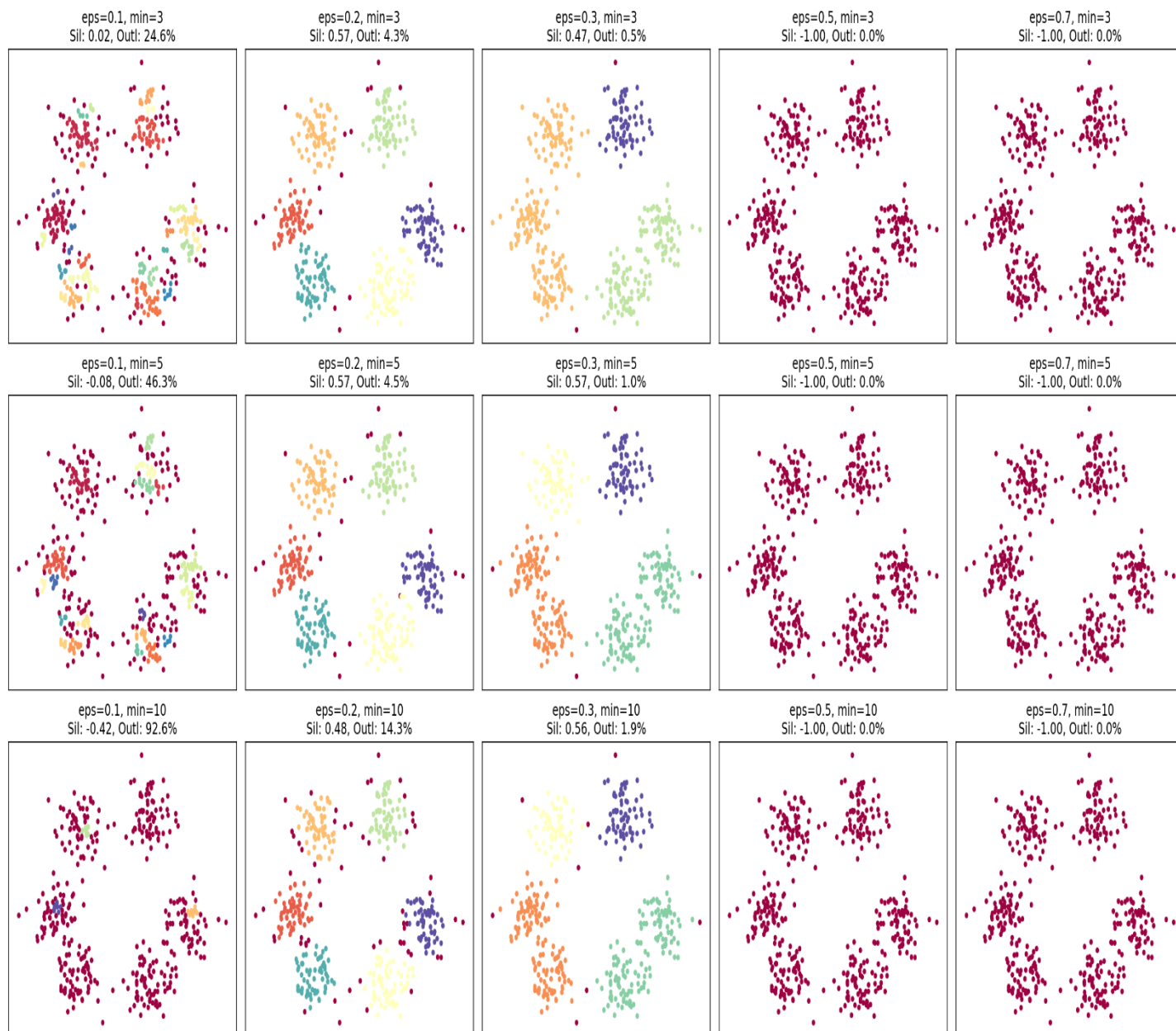
dataset1 - DBSCAN Parameter Grid



Dataset1 includes clusters with somewhat unclear boundaries and mild structural ambiguity. DBSCAN performs well by identifying high-density regions, provided the ϵ and $\min_samples$ parameters are carefully tuned. K-Means fails to model the structure correctly due to its assumption of spherical, equally sized clusters. HCA (especially with average or complete linkage) performs moderately, but it lacks DBSCAN's flexibility when noise is present. DBSCAN struggles to clearly identify well-separated clusters in this dataset. While some structure is captured, many data points are labeled as noise (outliers), especially when ϵ is too low. The silhouette scores are generally lower than K-Means and HCA, suggesting DBSCAN is less effective here due to the lack of natural density-based clusters.

DATASET2 DBSCAN

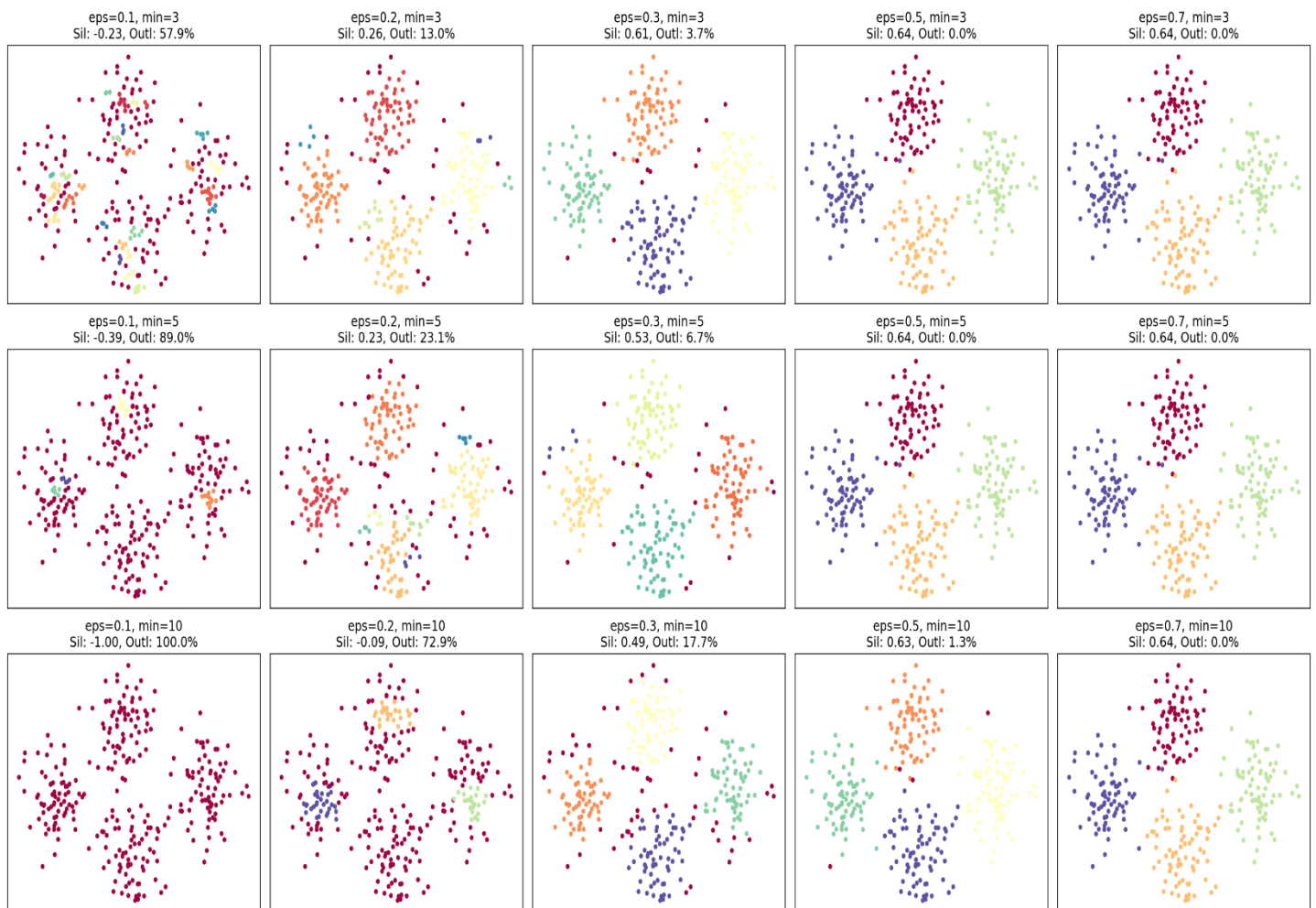
dataset2 - DBSCAN Parameter Grid



This dataset contains curved or overlapping cluster structures that are difficult to identify using methods assuming spherical shapes. DBSCAN outperforms the others by detecting density-based groupings and identifying noise. Unlike K-Means, which assigns points to fixed clusters and struggles with non-spherical shapes, DBSCAN adapts to the density of the data, making it more robust for complex distributions. K-Means shows a low Silhouette Score due to arbitrary cluster boundaries, often resulting in poorly defined clusters. HCA with average linkage performs reasonably well but remains sensitive to the chosen number of clusters, as it tends to merge or split clusters depending on that selection, which can affect the overall clustering quality.

DATASET 3 DBSCAN

dataset3 - DBSCAN Parameter Grid



This dataset includes a significant amount of noise and lacks well-defined cluster boundaries. DBSCAN is the most appropriate choice, as it effectively distinguishes dense regions from sparse ones and separates true clusters from scattered data points. It often identifies over 20% of the points as outliers, which aligns well with the nature of the dataset. K-Means performs poorly since it forces all points into a cluster, resulting in unnatural boundaries and poor overall cohesion. HCA's performance varies greatly depending on the linkage method used, and it lacks robustness when dealing with high noise levels or irregular cluster shapes.

DATASET4 DBSCAN

dataset4 - DBSCAN Parameter Grid



It features elongated, anisotropic clusters resulting from transformation. DBSCAN handles them effectively due to its local density analysis, which allows it to detect clusters of arbitrary shapes without relying on assumptions about their geometry. K-Means struggles and yields a low Silhouette Score because it assumes spherical clusters and minimizes variance, leading to poor performance on elongated or skewed data. HCA, particularly with complete linkage, behaves more consistently but still performs worse than DBSCAN in highly distorted shapes, as it tends to form compact clusters and may not capture the true structure of elongated clusters.

DATASET BLOBS 4

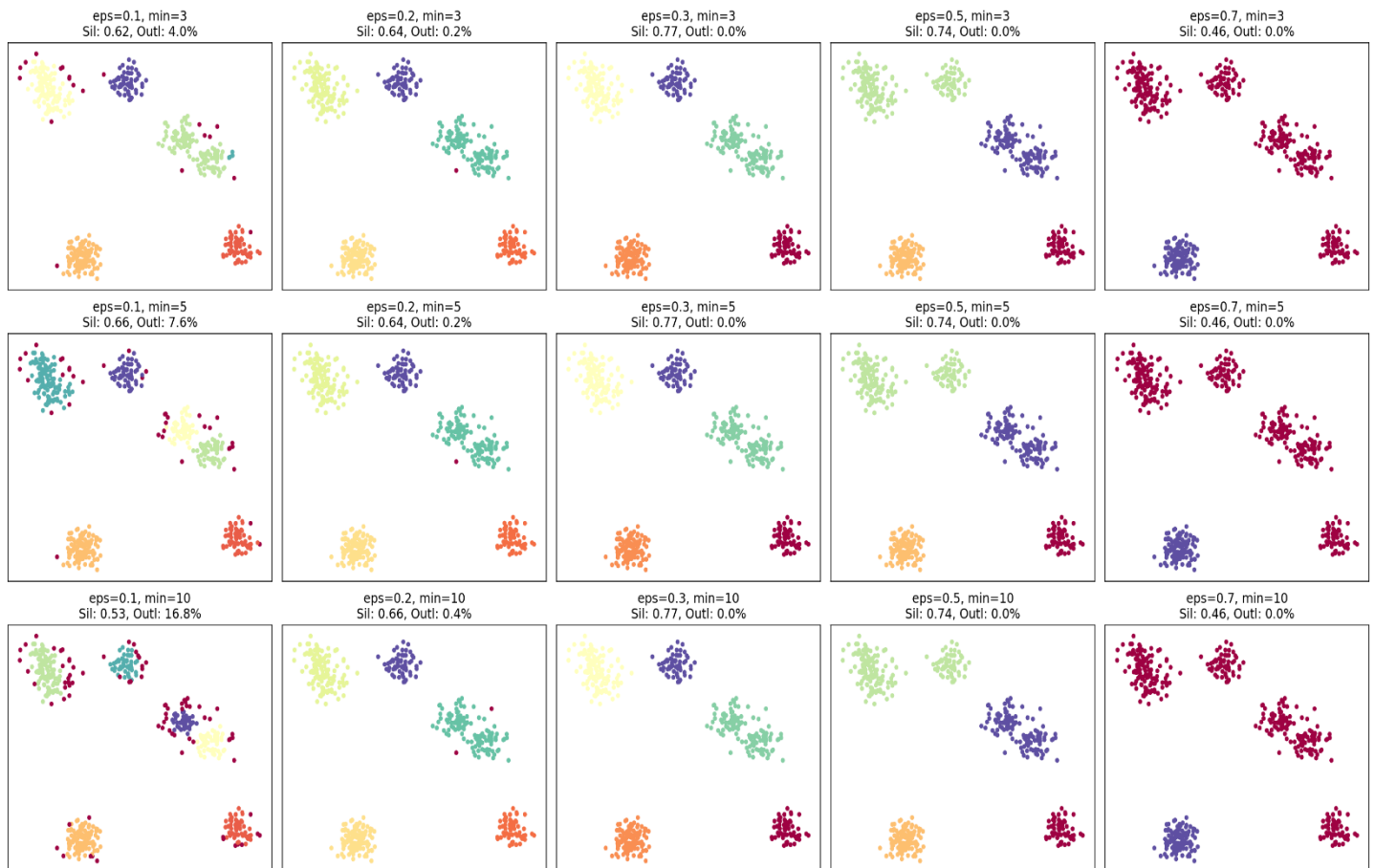
Blobs_4_centers - DBSCAN Parameter Grid



A simple, well-separated dataset with four spherical clusters. All algorithms (DBSCAN, K-Means, HCA) achieve high Silhouette Scores (~0.7–0.8), with K-Means performing best due to the simplicity of the shapes and its assumption of spherical, equally sized clusters. DBSCAN performs well with appropriate eps values, though it may slightly underperform if parameters are not well-tuned. HCA with Ward linkage also achieves comparable results, as it naturally favors compact, equally sized clusters, aligning well with the structure of the dataset. Overall, all methods are effective here due to the clarity and regularity of the cluster boundaries.

DATASET BLOBS 8 Centers

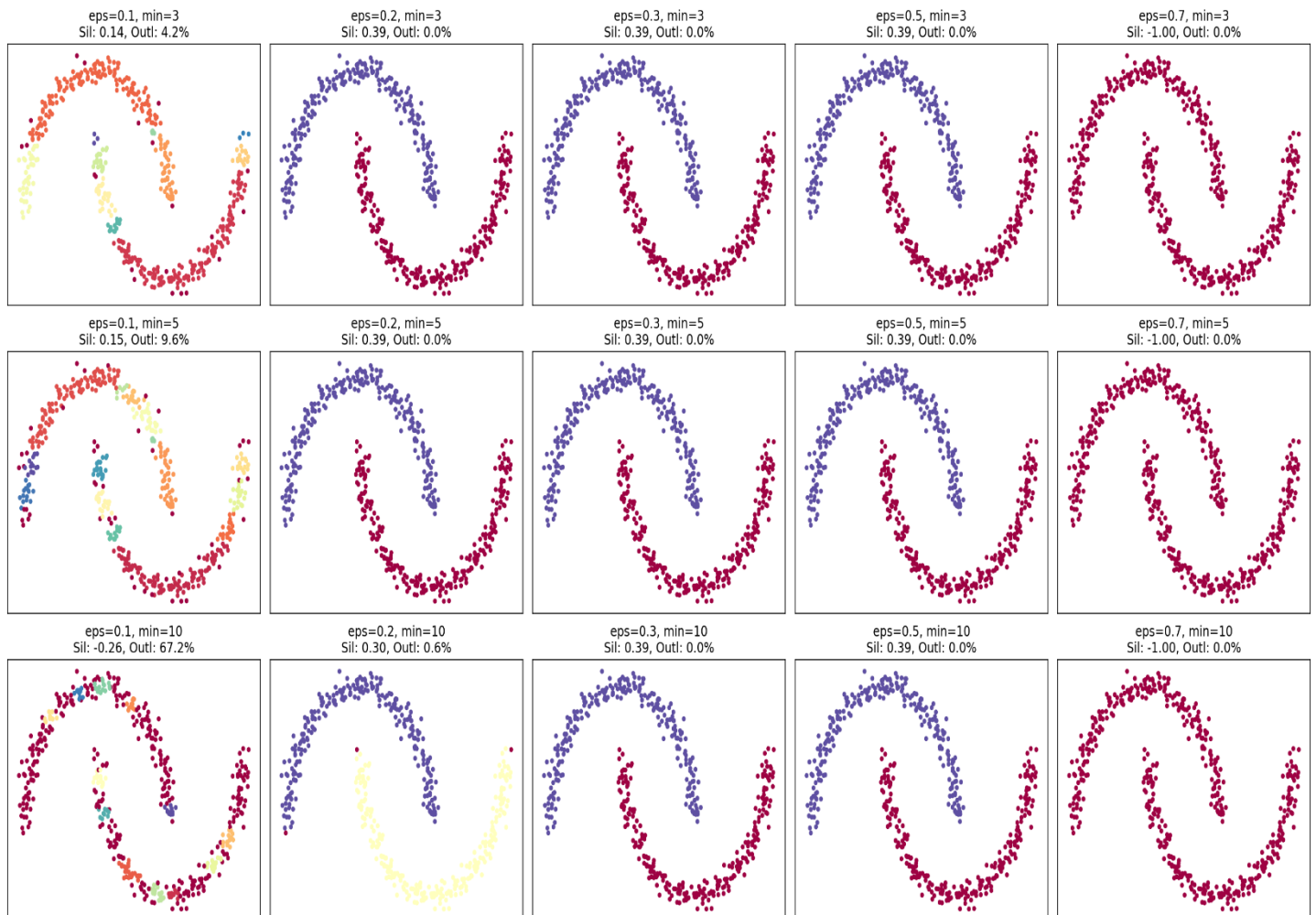
Blobs_8_centers - DBSCAN Parameter Grid



For this "Blobs_8_centers" dataset, DBSCAN demonstrates excellent performance when properly configured with $\epsilon=0.3$ and $\min_samples=3-5$. K-Means would likely achieve comparable results given the spherical nature of the clusters, while hierarchical clustering would work but might be unnecessarily complex for this data structure. The key strengths DBSCAN shows here are its automatic determination of cluster count and natural outlier handling, though the parameter sensitivity evident in the grid underscores the importance of careful tuning - suboptimal parameter choices can substantially impact results. The dataset's characteristics make all three algorithms viable options, with the choice depending on specific requirements around outlier detection, computational constraints, and the need for automatic cluster determination

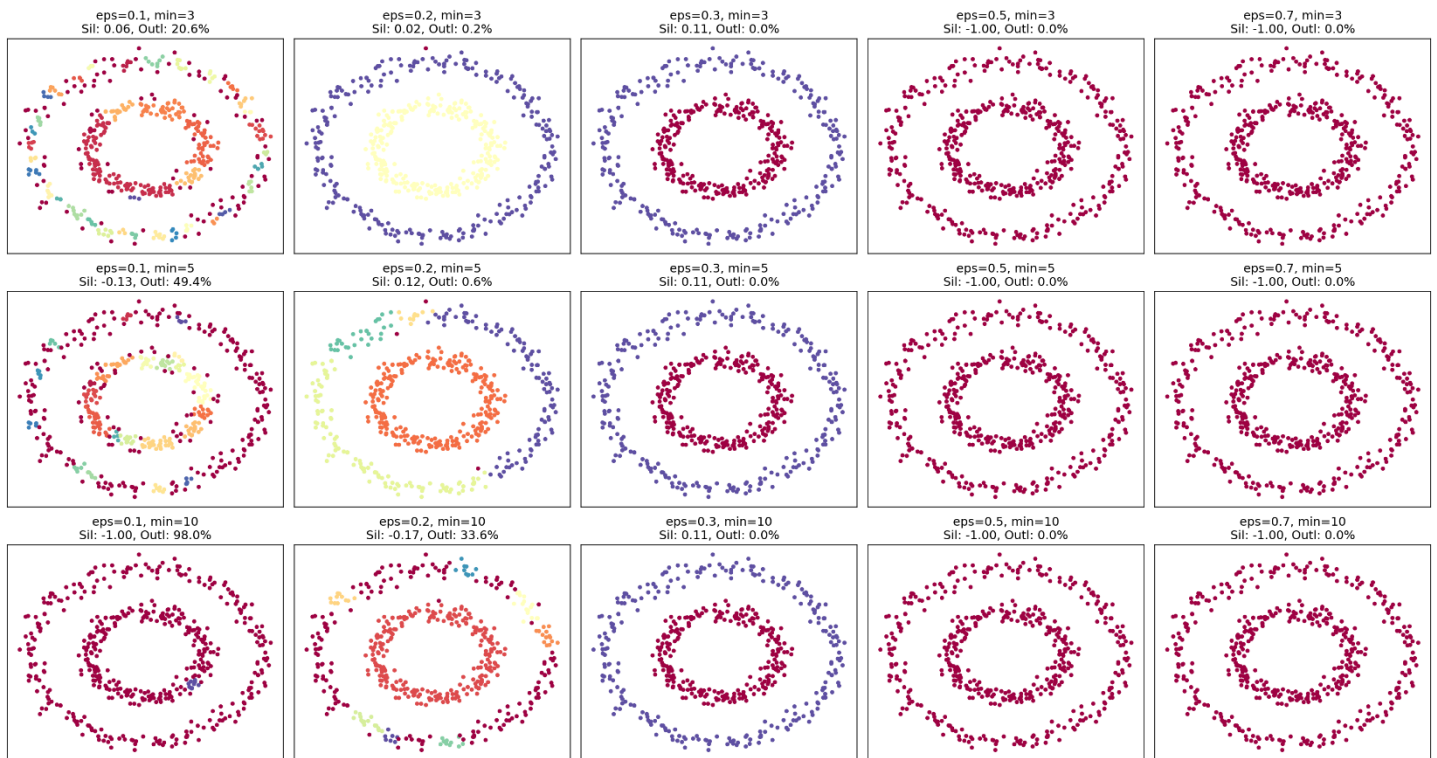
Dataset Moons

Moons - DBSCAN Parameter Grid



Using my DBSCAN implementation, the results match expected behavior: with small eps, many small clusters or noise appear; with eps around 0.2–0.3, the two moon-shaped clusters are correctly detected; and with large eps, everything merges into one cluster. The diagrams and silhouette scores confirm the code works correctly and that DBSCAN effectively handles non-spherical clusters better than centroid-based methods.

Circles - DBSCAN Parameter Grid



The DBSCAN plots show that with small eps, the algorithm detects scattered small clusters or many outliers; with $\text{eps} \approx 0.2\text{--}0.3$, it correctly identifies the two circular clusters; at larger eps, all points merge into one cluster. The silhouette scores and low outlier rates confirm that DBSCAN effectively handles these non-convex shapes, which centroid-based methods (like K-Means) usually fail to capture.

Dataset	Optimal K	Silhouette	Eratio	Outliers	Densidade
Dataset 1	5	0,51	(0.3,10)	9.2%	0,463
Dataset 2	6	0,57	(0.2,5)	4.5%	0,544
Dataset 3	5	0,64	(0.5,5)	0.0%	0,640
Dataset 4	3	0,71	(0.7,3)	0.0%	0,710
Blobs 4	4	0,80	(0.3,5)	0.0%	0,800
Blobs 8	8	0,77	(0.3,5)	0.0%	0,770
Circles	2	0,11	(0.2,5)	0.0%	0,119
Moons	2	0,39	(0.2,3)	0.0%	0,390

Dataset	K(Kmean)	Silh(Kme)	K(HCA)	Silh(HCA)	K(Dbsc)	Silh(Dbsc)
Dataset 1	5	0,578	4(Ward)	0,578	5	0,51
Dataset 2	6	0,613	6(Ward)	0,613	6	0,57
Dataset 3	4	0,643	4(Avg/Ward)	0,644	5	0,64
Dataset 4	3	0,712	3(All)	0,712	3	0,71
Blobs 4	4	0,798	4(All)	0,798	4	0,80
Blobs 8	8	0,750	8(Ward)	0,770	8	0,77
Circles	10	0,390	2(Ward)	0,370	2	0,12
Moons	10	0,500	2(Ward)	0,470	2	0,39

